

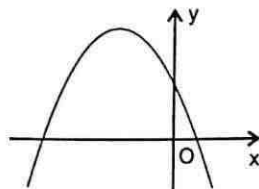
49. If the roots α, β of the equation $px^2 + qx + r = 0$ are real and of opposite sign (where p, q, r are real coefficient), then the roots of the equation $\alpha(x - \beta)^2 + \beta(x - \alpha)^2 = 0$ are:
 (a) positive (b) negative
 (c) real and of opposite sign (d) imaginary
50. Let a, b and c be three distinct real roots of the cubic $x^3 + 2x^2 - 4x - 4 = 0$.
 If the equation $x^3 + qx^2 + rx + s = 0$ has roots $\frac{1}{a}, \frac{1}{b}$ and $\frac{1}{c}$, then the value of $(q + r + s)$ is equal to :
 (a) $\frac{3}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $\frac{1}{6}$
51. Solution set of the inequality, $2 - \log_2(x^2 + 3x) \geq 0$ is :
 (a) $[-4, 1]$ (b) $[-4, -3) \cup (0, 1]$ (c) $(-\infty, -3) \cup (1, \infty)$ (d) $(-\infty, -4) \cup [1, \infty)$
52. For what least integral 'k' is the quadratic trinomial $(k - 2)x^2 + 8x + (k + 4)$ is positive for all real values of x ?
 (a) $k = 4$ (b) $k = 5$ (c) $k = 3$ (d) $k = 6$
53. If roots of the equation $(m - 2)x^2 - (8 - 2m)x - (8 - 3m) = 0$ are opposite in sign, then number of integral values(s) of m is/are :
 (a) 0 (b) 1 (c) 2 (d) more than 2
54. If $\log_{0.6} \left(\log_6 \left(\frac{x^2 + x}{x + 4} \right) \right) < 0$, then complete set of value of 'x' is :
 (a) $(-4, -3) \cup (8, \infty)$ (b) $(-\infty, -3) \cup (8, \infty)$
 (c) $(8, \infty)$ (d) None of these
55. Two different real numbers α and β are the roots of the quadratic equation $ax^2 + c = 0$ with $a, c \neq 0$, then $\alpha^3 + \beta^3$ is :
 (a) a (b) $-c$ (c) 0 (d) -1
56. The least integral value of 'k' for which $(k - 1)x^2 - (k + 1)x + (k + 1)$ is positive for all real value of x is:
 (a) 1 (b) 2 (c) 3 (d) 4
57. If $(-2, 7)$ is the highest point on the graph of $y = -2x^2 - 4ax + k$, then k equals:
 (a) 31 (b) 11 (c) -1 (d) $-1/3$
58. If $a + b + c = 0$, $a, b, c \in Q$ then roots of the equation $(b + c - a)x^2 + (c + a - b)x + (a + b - c) = 0$ are :
 (a) rational (b) irrational (c) imaginary (d) none of these
59. If two roots of $x^3 - ax^2 + bx - c = 0$ are equal in magnitude but opposite in sign. Then:
 (a) $a + bc = 0$ (b) $a^2 = bc$
 (c) $ab = c$ (d) $a - b + c = 0$

60. If α and β are the real roots of $x^2 + px + q = 0$ and α^4, β^4 are the roots of $x^2 - rx + s = 0$. Then the equation $x^2 - 4qx + 2q^2 - r = 0$ has always ($\alpha \neq \beta, p \neq 0, p, q, r, s \in R$):
 (a) one positive and one negative root (b) two positive roots
 (c) two negative roots (d) can't say anything
61. If $x^2 + px + 1$ is a factor of $ax^3 + bx + c$, then:
 (a) $a^2 + c^2 = -ab$ (b) $a^2 + c^2 = ab$ (c) $a^2 - c^2 = ab$ (d) $a^2 - c^2 = -ab$
62. In a $\triangle ABC$ $\tan \frac{A}{2}, \tan \frac{B}{2}, \tan \frac{C}{2}$ are in H.P., then the value of $\cot \frac{A}{2} \cot \frac{C}{2}$ is :
 (a) 3 (b) 2 (c) 1 (d) $\sqrt{3}$
63. Let $f(x) = 10 - |x - 10| \forall x \in [-9, 9]$, if M and m be the maximum and minimum value of $f(x)$ respectively, then :
 (a) $M + m = 0$ (b) $2M + m = -9$ (c) $2M + m = 7$ (d) $M + m = 7$
64. Solution of the quadratic equation $(3|x| - 3)^2 = |x| + 7$, which belongs to the domain of the function $y = \sqrt{(x - 4)x}$ is :
 (a) $\pm \frac{1}{9}, \pm 2$ (b) $\frac{1}{9}, 8$ (c) $-2, -\frac{1}{9}$ (d) $-\frac{1}{9}, 8$
65. Number of real solutions of the equation $x^2 + 3|x| + 2 = 0$ is :
 (a) 0 (b) 1 (c) 2 (d) 4
66. If the roots of equation $x^2 - bx + c = 0$ be two consecutive integers, then $b^2 - 4c =$
 (a) 3 (b) -2 (c) 1 (d) 2
67. If x is real, then maximum value of $\frac{3x^2 + 9x + 17}{3x^2 + 9x + 7}$ is :
 (a) 41 (b) 1 (c) $\frac{17}{7}$ (d) $\frac{1}{4}$
68. If $\frac{x^2 + 2x + 7}{2x + 3} < 6, x \in R$ then :
 (a) $x \in \left(-\infty, -\frac{3}{2}\right) \cup (11, \infty)$ (b) $x \in (-\infty, -1) \cup (11, \infty)$
 (c) $x \in \left(-\frac{3}{2}, -1\right)$ (d) $x \in \left(-\infty, -\frac{3}{2}\right) \cup (-1, 11)$
69. If x is real, then range of $\frac{3x - 2}{7x + 5}$ is :
 (a) $R - \left\{\frac{2}{5}\right\}$ (b) $R - \left\{\frac{3}{7}\right\}$ (c) $(-\infty, \infty)$ (d) $R - \left\{\frac{-2}{5}\right\}$
70. Let A denotes the set of values of x for which $\frac{x + 2}{x - 4} \leq 0$ and B denotes the set of values of x for which $x^2 - ax - 4 \leq 0$. If B is the subset of A , then a **CAN NOT** take the integral value :
 (a) 0 (b) 1 (c) 2 (d) 3

71. If the quadratic polynomial $P(x) = (p-3)x^2 - 2px + 3p - 6$ ranges from $[0, \infty)$ for every $x \in \mathbb{R}$, then the value of p can be :

(a) 3 (b) 4 (c) 6 (d) 7

72. If graph of the quadratic $y = ax^2 + bx + c$ is given below :



then :

(a) $a < 0, b > 0, c > 0$ (b) $a < 0, b > 0, c < 0$
 (c) $a < 0, b < 0, c > 0$ (d) $a < 0, b < 0, c < 0$

73. If quadratic equation $ax^2 + bx + c = 0$ does not have real roots, then which of the following may be false :

(a) $a(a-b+c) > 0$ (b) $c(a-b+c) > 0$
 (c) $b(a-b+c) > 0$ (d) $(a+b+c)(a-b+c) > 0$

74. Minimum value of $y = x^2 - 3x + 5, x \in [-4, 1]$ is :

(a) 3 (b) $\frac{11}{4}$ (c) 0 (d) 9

75. If $3x^2 - 17x + 10 = 0$ and $x^2 - 5x + m = 0$ has a common root, then sum of all possible real values of 'm' is :

(a) 0 (b) $-\frac{26}{9}$ (c) $\frac{29}{9}$ (d) $\frac{26}{3}$

76. For real numbers x and y , if $x^2 + xy - y^2 + 2x - y + 1 = 0$, then :

(a) y can not be between 0 and $\frac{8}{5}$ (b) y can not be between $-\frac{8}{5}$ and $\frac{8}{5}$
 (c) y can not be between $-\frac{8}{5}$ and 0 (d) y can not be between $-\frac{16}{5}$ and 0

77. If $3x^4 - 6x^3 + kx^2 - 8x - 12$ is divisible by $x - 3$, then it is also divisible by :

(a) $3x^2 - 4$ (b) $3x^2 + 4$
 (c) $3x^2 + x$ (d) $3x^2 - x$

78. The complete set of values of a so that equation $\sin^4 x + a \sin^2 x + 4 = 0$ has at least one real root is :

(a) $(\infty, -5]$ (b) $(-\infty, 4] \cup [4, \infty)$
 (c) $(-\infty, -4]$ (d) $[4, \infty)$

79. Let r, s, t be the roots of the equation $x^3 + ax^2 + bx + c = 0$, such that $(rs)^2 + (st)^2 + (rt)^2 = b^2 - kac$, then $k =$

(a) 1 (b) 2 (c) 3 (d) 4

80. If the roots of the cubic $x^3 + ax^2 + bx + c = 0$ are three consecutive positive integers, then the value of $\frac{a^2}{b+1} =$
- (a) 1 (b) 2 (c) 3 (d) 4
81. Let 'k' be a real number. The minimum number of distinct real roots possible of the equation $(3x^2 + kx + 3)(x^2 + kx - 1) = 0$ is :
- (a) 0 (b) 2 (c) 3 (d) 4
82. If r and s are variables satisfying the equation $\frac{1}{r+s} = \frac{1}{r} + \frac{1}{s}$. The value of $\left(\frac{r}{s}\right)^3$ is equal to :
- (a) 1 (b) -1
(c) 3 (d) not possible to determine
83. Let $f(x) = x^2 + ax + b$. If the maximum and the minimum values of $f(x)$ are 3 and 2 respectively for $0 \leq x \leq 2$, then the possible ordered pair of (a, b) is :
- (a) $(-2, 3)$ (b) $(-3/2, 2)$ (c) $(-5/2, 3)$ (d) $(-5/2, 2)$
84. The roots of the equation $|x^2 - x - 6| = x + 2$ are given by :
- (a) -2, 2, 4 (b) 0, 1, 4 (c) -2, 1, 4 (d) 0, 2, 4
85. If a, b, c be the sides of $\triangle ABC$ and equations $ax^2 + bx + c = 0$ and $5x^2 + 12x + 13 = 0$ have a common root, then $\angle C$ is :
- (a) 60° (b) 90° (c) 120° (d) 45°
86. If α, β and γ are three real roots of the equation $x^3 - 6x^2 + 5x - 1 = 0$, then the value of $\alpha^4 + \beta^4 + \gamma^4$ is :
- (a) 250 (b) 650 (c) 150 (d) 950
87. If one of the roots of the equation $2x^2 - 6x + k = 0$ is $\frac{\alpha + 5i}{2}$, then the value of α and k are :
- (a) $\alpha = 3, k = 8$ (b) $\alpha = \frac{3}{2}, k = 17$ (c) $\alpha = -3, k = -17$ (d) $\alpha = 3, k = 17$
88. Let x_1 and x_2 be the real roots of the equation $x^2 - (k-2)x + (k^2 + 3k + 5) = 0$, then the maximum value of $x_1^2 + x_2^2$ is :
- (a) 19 (b) 18 (c) $\frac{50}{9}$ (d) non-existent
89. The complete set of values of 'a' for which the inequality $(a-1)x^2 - (a+1)x + (a-1) \geq 0$ is true for all $x \geq 2$.
- (a) $\left[\frac{3}{7}, 1\right]$ (b) $(-\infty, 1)$ (c) $\left(-\infty, \frac{7}{3}\right]$ (d) $\left[\frac{7}{3}, \infty\right)$
90. If α, β be the roots of $4x^2 - 17x + \lambda = 0, \lambda \in R$ such that $1 < \alpha < 2$ and $2 < \beta < 3$, then the number of integral values of λ is :
- (a) 1 (b) 2 (c) 3 (d) 4

91. Assume that p is a real number. In order of $\sqrt[3]{x+3p+1} - \sqrt[3]{x} = 1$ to have real solutions, it is necessary that:
 (a) $p \geq 1/4$ (b) $p \geq -1/4$ (c) $p \geq 1/3$ (d) $p \geq -1/3$
92. If α, β are the roots of the quadratic equation $x^2 - \left(3 + 2^{\sqrt{\log_2 3}} - 3^{\sqrt{\log_3 2}}\right)x - 2(3^{\log_3 2} - 2^{\log_2 3}) = 0$, then the value of $\alpha^2 + \alpha\beta + \beta^2$ is equal to :
 (a) 3 (b) 5 (c) 7 (d) 11
93. The minimum value of $f(x, y) = x^2 - 4x + y^2 + 6y$ when x and y are subjected to the restrictions $0 \leq x \leq 1$ and $0 \leq y \leq 1$, is :
 (a) -1 (b) -2 (c) -3 (d) -5
94. The expression $ax^2 + 2bx + c$, where ' a ' is non-zero real number, has same sign as that of ' a ' for every real value of x , then roots of quadratic equation $ax^2 + (b-c)x - 2b - c - a = 0$, are :
 (a) real and equal (b) real and unequal
 (c) non-real having positive real part (d) non-real having negative real part
95. Let a, b and c be the roots of $x^3 - x + 1 = 0$, then the value of $\left(\frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1}\right)$ equals to :
 (a) 1 (b) -1 (c) 2 (d) -2
96. The number of integral values of k for which the inequality $x^2 - 2(4k-1)x + 15k^2 - 2k - 7 \geq 0$ holds for all $x \in R$ is :
 (a) 2 (b) 3 (c) 4 (d) infinite
97. The number of integral values which can be taken by the expression, $f(x) = \frac{x^3 - 1}{(x-1)(x^2 - x + 1)}$ for $x \in R$, is :
 (a) 1 (b) 2 (c) 3 (d) infinite
98. The complete set of values of m for which the inequality $\frac{x^2 - mx - 2}{x^2 + mx + 4} > -1$ is satisfied $\forall x \in R$, is :
 (a) $m = 0$ (b) $-1 < m < 1$ (c) $-2 < m < 2$ (d) $-4 < m < 4$
99. The complete set of values of a for which the roots of the equation $x^2 - 2|a+1|x + 1 = 0$ are real is given by :
 (a) $(-\infty, -2] \cup [0, \infty)$ (b) $(-\infty, -1] \cup [0, \infty)$
 (c) $(-\infty, -1] \cup [1, \infty)$ (d) $(-\infty, -2] \cup [1, \infty)$
100. The quadratic polynomials defined on real coefficients $P(x) = a_1x^2 + 2b_1x + c_1$, $Q(x) = a_2x^2 + 2b_2x + c_2$, $P(x)$ and $Q(x)$ both take positive values $\forall x \in R$. If $f(x) = a_1a_2x^2 + b_1b_2x + c_1c_2$, then :
 (a) $f(x) < 0 \forall x \in R$
 (b) $f(x) > 0 \forall x \in R$